Top-K Off-Policy Correction for a REINFORCE Recommender System https://arxiv.org/pdf/1812.02353

MDP: $(S, \mathcal{A}, P, R, \rho_0, \gamma)$

$\pi(a|s)$ – ?

$$\max_{\pi} J(\pi) = \mathbb{E}_{\tau\sim\pi}[R(\tau)], \quad R(\tau) = \sum_{t=0}^{|\tau|} r(s_t, a_t)\gamma^t$$

$\tau = (s_0, a_0, s_1, a_1, \ldots)$, $s_0 \sim \rho_0$

$$J(\pi_\theta) = \mathbb{E}_{s_0\sim\rho_0, a_t\sim\pi_\theta(\cdot|s_t), s_{t+1}\sim P(\cdot|s_t,a_t)}\left[\sum_{t=0}^{|\tau|} r(s_t, a_t)\gamma^t\right]$$

$$\nabla_\theta J(\pi_\theta) = \nabla_\theta \sum_\tau p_\theta(\tau) R(\tau) = \sum_\tau \nabla_\theta p_\theta(\tau) R(\tau) =$$

$$= \sum_\tau p_\theta(\tau)\frac{\nabla_\theta p_\theta(\tau)}{p_\theta(\tau)} R(\tau) = \sum_\tau p_\theta(\tau)\nabla_\theta \log p_\theta(\tau) R(\tau) =$$

$$= \mathbb{E}_\tau\left[\nabla_\theta \log p_\theta(\tau) R(\tau)\right] = \mathbb{E}_\tau\left[\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot\right.$$

$$\left. \times R(\tau)\right] \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_{it}|s_{it}) \cdot R(\tau_i)$$

Behaviour clonning:

$$J_{BC}(\pi) = \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{|\tau|} \log \pi_\theta(a_t|s_t)\right] \Rightarrow$$

$$\Rightarrow \nabla J_{BC}(\pi) = \mathbb{E}_{\tau\sim D}\left[\sum_{t=0}^{|\tau|} \log \pi_\theta(a_t|s_t) \cdot 1\right]$$

$= B_t$

$$\nabla_\theta J(\pi_\theta) = \underbrace{\mathbb{E}\left[\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{t'<t} \gamma^{t'} r_{t'}\right)\right]}_{(1)} +$$

$$+ \mathbb{E}\left[\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{t' \geq t} \gamma^{t'} r_{t'}\right)\right] =$$

$$(1) = \sum_{t=0}^{\infty} \mathbb{E}\, \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot B_t = \sum_{t=0}^{\infty} \mathbb{E}\Big[B_t \times$$

$$\times\, \mathbb{E}\underbrace{\left[\nabla_\theta \log \pi_\theta(a_t|s_t)|s_t\right]}\Big] = 0$$

$$\sum_a \pi_\theta(a_t|s_t)\nabla_\theta \log \ldots = \sum_a \nabla_\theta \pi_\theta(a_t|s_t) = \nabla 1 = 0$$

$$= \mathbb{E}\left[\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_t|s_t)\, \gamma^t G_t\right] =$$

return-to-go

$d_t^\pi(\cdot)$ – (discounted) state visitation frequency at time t under policy π

$$= \mathbb{E}_\tau\left[\sum_{t=0}^{|\tau|} \nabla_\theta \log \pi_\theta(a_t|s_t)\, \gamma^t R_t(s_t,a_t)\right] =$$

$$R_t(s_t,a_t) = \sum_{t'=t}^{|\tau|} \gamma^{t'-t} r(s_{t'},a_{t'})$$

$$d_\gamma^\pi(s) = \sum_{t=0}^{\infty} \gamma^t \Pr(s_t = s|\pi)$$

$$d_t^\pi(s) = Pr(s_t = s \mid \pi)$$

$$\ominus \sum_{t=0}^{\infty} \mathbb{E}\left[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \gamma^t R_t(s_t, a_t)\right] =$$

$$= \sum_{t=0}^{\infty} \mathbb{E}_{s_t \sim d_t^\pi(\cdot),\, a_t \sim \pi(\cdot \mid s_t)} \left[R_t(s_t, a_t)\, \nabla_\theta \log \pi_\theta(a_t \mid s_t) \cdot \gamma^t\right]$$

By convention

$$\rightarrow \sum_{t=0}^{\infty} \mathbb{E}_{s_t \sim d_t^\pi(\cdot),\, a_t \sim \pi(\cdot \mid s_t)} \left[R_t(s_t, a_t)\, \nabla_\theta \log \pi_\theta(a_t \mid s_t)\right] =$$

β - Behavior policy

$$= \sum_{t=0}^{\infty} \mathbb{E}_{s_t \sim d_t^\beta(\cdot),\, a_t \sim \beta(\cdot \mid s_t)} \Bigg[\frac{d_t^\pi(s_t)}{d_t^\beta(s_t)} \cdot \frac{\pi_\theta(a_t \mid s_t)}{\beta(a_t \mid s_t)} \times$$

$$\times \prod_{t'=t+1}^{|\tau|} \frac{\pi_\theta(a_{t'} \mid s_{t'})}{\beta(a_{t'} \mid s_{t'})} \cdot \nabla_\theta \log \pi_\theta(a_t \mid s_t) \Bigg]$$

$$w(s_t, a_t) = \frac{d_t^\pi(s_t)}{d_t^\beta(s_t)} \cdot \prod_{t' \leq t}^{|\tau|} \frac{\pi_\theta(a_{t'} \mid s_{t'})}{\beta(a_{t'} \mid s_{t'})}$$ $\leftarrow$ importance weight

$$\nabla_\theta J(\pi_\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{|\tau_i|} w_t \cdot \nabla_\theta \log \pi_\theta(a_t \mid s_t) \cdot R_t$$

$$\nabla_\theta J_{BC}(\pi_\theta) \approx \frac{1}{D} \sum_{(s_t, a_t) \in D} \nabla_\theta \log \pi_\theta(a_t \mid s_t)$$

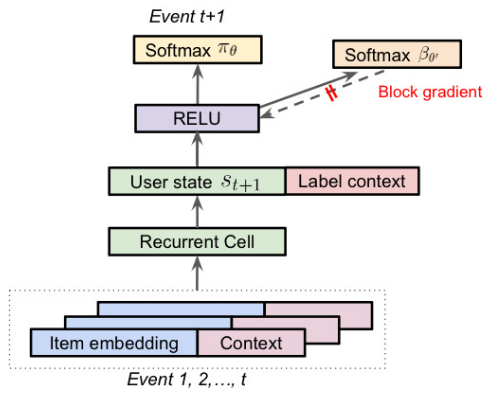

Figure 1: A diagram shows the parametrisation of the policy π_θ as well as the behavior policy $\beta_{\theta'}$.

Top-K Off-Policy Correction

$\Pi_\theta(A|s)$ - ?

$$\max_\theta J(\Pi_\theta) = \mathbb{E}_{s_t \sim d_t^\Pi(\cdot),\, A_t \sim \Pi_\theta(\cdot|s_t)} \left[R_t(s_t, A_t) \right]$$

Simplification:

$$\alpha_\theta(a|s) = 1 - (1 - \pi_\theta(a|s))^k$$

REINFORCE Off-Policy Correction Enhancing for Recommender Systems